

Homework (Carolina Reaper)

- Q1.** A music producer has x hours to record a song and y hours to mix it. If $x + 2y \leq 10$ and $x \geq 2$, what is the maximum value of y ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** An artist is creating a painting with a canvas size of x feet by y feet. If $x \geq 3$, $y \geq 2$, and $2x + y \leq 12$, what is the maximum area of the painting?
- (A) 18 sq ft
(B) 20 sq ft
(C) 24 sq ft
(D) 30 sq ft
(E) 36 sq ft
- Q3.** A rhythm pattern has a frequency of x beats per minute and a duration of y minutes. If $x \geq 100$, $y \geq 2$, and $x \cdot y \leq 1200$, what is the maximum duration of the rhythm pattern?
- (A) 8 min
(B) 10 min
(C) 12 min
(D) 15 min
(E) 20 min
- Q4.** A music festival has x stages and y food vendors. If $x + y \leq 15$ and $x \geq 3$, what is the maximum number of food vendors?
- (A) 8
(B) 10
(C) 12
(D) 15
(E) 20
- Q5.** An arts center has a budget of _____
 x
- _____ for a new exhibit and _____
 y
- _____ for security. If $x + 2y \leq 50000$ and $x \geq 10000$, find the maximum amount for security and the corresponding exhibit budget.
- Q6.** A graphic designer is creating a poster with a width of x inches and a height of y inches. If $x \geq 10$, $y \geq 5$, and $2x + y \leq 30$, what is the maximum area of the poster?
- (A) 50 sq in
(B) 60 sq in
(C) 75 sq in
(D) 100 sq in
(E) 120 sq in
- Q7.** A music school has x students and y teachers. If $x \geq 20$, $y \geq 5$, and $x \cdot y \leq 200$, what is the maximum number of teachers?
- (A) 5
(B) 8
(C) 10
(D) 12
(E) 15
- Q8.** A concert has x seats and y concession stands. If $x + 2y \leq 1000$ and $x \geq 500$, what is the maximum number of concession stands?
- (A) 200
(B) 250
(C) 300
(D) 400
(E) 500

Open Ended Questions (FRQ)

- Q9.** A museum has a budget of _____
 x
- _____ for a new exhibit and _____
 y
- _____ for security. If $x + 2y \leq 50000$ and $x \geq 10000$, find the maximum amount for security and the corresponding exhibit budget.
- Q10.** A recording studio has x hours to record a song and y hours to mix it. If $x + 2y \leq 20$ and $x \geq 4$, find the maximum value of y and the corresponding recording time.